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CG-Reasoner: Centroid-Guided Positional Reasoning Segmentation for Medical Imaging with a Robust Visual-Text Consistency Metric

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Abstract

Accurate and interpretable medical image segmentation remains a major challenge, as existing deep learning models primarily optimize pixel-level accuracy while overlooking positional reasoning—an essential component for automated report generation and clinical interpretability. We introduce CG-Reasoner, a novel centroid-guided cross-modal framework that jointly performs medical image segmentation and positional reasoning. CG-Reasoner integrates a multimodal large language model (LLM), a newly designed light-weight encoder–decoder architecture, and a Text2Centroid module that predicts lesion centroids from reasoning embeddings—enabling the model to produce both accurate segmentation masks and spatially coherent, clinically meaningful reasoning explanations. Furthermore, we propose PRScore (Positional-Reasoning Score), a robust evaluation metric that jointly measures the spatial and semantic alignment between generated reasoning text and segmentation masks. Experiments on six medical datasets across different imaging modalities demonstrate that CG-Reasoner achieves state-of-the-art performance, offering precise segmentation, spatially coherent reasoning, and clinically interpretable visual-textual explanations within a unified framework. The source code is available at https://github.com/lpmm2025/CG-Reasoner.

1. Introduction

Medical image segmentation plays a central role in clinical workflows, serving as a foundation for diagnosis, treatment planning, and disease monitoring. While modern deep learning models have achieved impressive segmentation accuracy, they primarily focus on pixel-level delineation and lack the ability to reason about spatial relationships or describe lesion locations in human-interpretable terms. In clinical practice, however, radiologists rely not only on visual segmentation but also on spatial reasoning—describing where a lesion appears relative to surrounding anatomy. Integrating such positional reasoning into segmentation models is critical for developing AI systems that can automatically generate diagnostic narratives, streamline report writing, and assist clinicians in decision-making.

Despite recent advances in vision–language models (VLMs) and multimodal large language models (LLMs), existing medical segmentation frameworks typically treat reasoning and segmentation as separate tasks. Vision modules produce masks, while language modules generate generic textual summaries, leading to poor spatial consistency between the reasoning and the actual lesion location. Moreover, current evaluation metrics (e.g., Dice or IoU) measure pixel overlap but fail to assess whether the model's reasoning text correctly localizes the region of interest. To address these limitations, we present CG-Reasoner, a centroid-guided multimodal framework that unifies segmentation and positional reasoning. Fig. 1 shows the prompts and the corresponding responses generated by our CG-Reasoner. By coupling a Text2Centroid model with a multimodal LLM, CG-Reasoner explicitly grounds textual reasoning in spatial geometry, resulting in more interpretable and spatially coherent medical image analysis. Furthermore, we propose PRScore (Positional-Reasoning Score), a new evaluation metric that quantifies the alignment between generated reasoning text and spatial mask predictions, enabling interpretable benchmarking of multimodal segmentation–reasoning systems. Our key contributions are summarized as follows:

- We propose CG-Reasoner, a unified cross-modal framework that jointly performs segmentation and positional reasoning. It enables the multimodal LLM to reason about lesion location while the vision encoder and decoder collaboratively generate spatially grounded segmentation output.
- We develop a novel, light-weight encoder–decoder design that jointly integrates multi-scale visual refinement with adaptive cross-modal fusion. The encoder efficiently captures fine-grained anatomical details across di-

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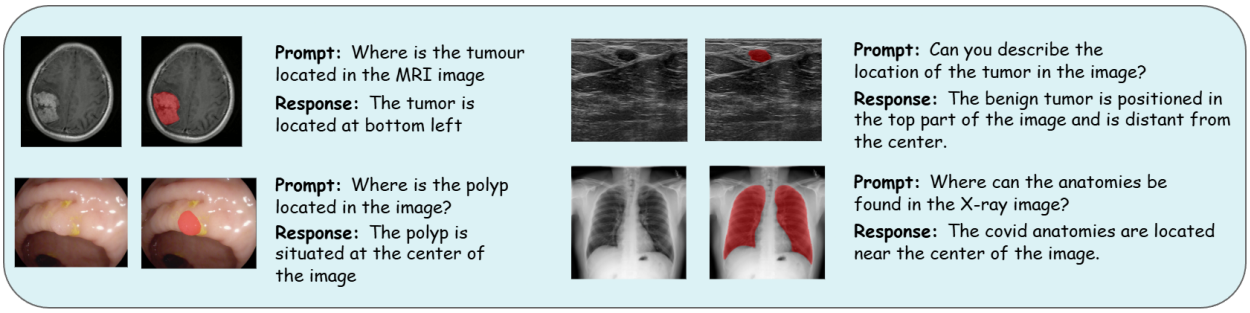


Figure 1. Examples of visual reasoning outputs generated by CG-Reasoner across a variety of medical images and imaging modalities.

verse imaging modalities, while the decoder employs our custom semantic-guided fusion mechanism to align visual and linguistic representations for context-aware and boundary-precise segmentation, all with minimal computational overhead.

- We propose a Text2Centroid (T2C) module, which predicts lesion centroids directly from reasoning text to provide explicit spatial grounding for multimodal explanations. Alongside this, we introduce PRScore, a metric that jointly evaluates spatial and semantic alignment between reasoning text and segmentation masks, offering a more faithful and interpretable assessment of multimodal reasoning and localization.
- We evaluate our CG-Reasoner across six datasets spanning multiple medical imaging modalities and demonstrate that it outperforms existing state-of-the-art methods in both segmentation and reasoning tasks.

2. Related Work

2.1. Medical Image Segmentation

In the medical domain, segmentation has been dominated by fully supervised convolutional- and transformer-based architectures. Examples include the canonical U–Net [17], ResUNet++ [6], nnU–Net [5], DoubleUNet [7], TransResUNet [25], and Swin–UNet [2]. More recently, promptable frameworks adapted from foundation segmentation models, such as *MedSAM* [12] and *SAM-Med2D* [4], allow point/box prompting for medical images. These methods, however, lack rich integration with free-form language queries or positional reasoning. The recent system *BiomedParse*[24] leverages text prompts to infer object shapes and positions in biomedical images, but it still does not offer a fully interactive segmentation system with textual positional cues, nor does it tie reasoning and segmentation in a unified framework. By contrast, our proposed method explicitly couples positional reasoning (via language) with segmentation mask generation in a medical context, thereby filling an important gap in interactive, position-aware medical image segmentation.

2.2. Language-Guided Reasoning Segmentation

Recent advancements in combining segmentation with language-based reasoning have introduced tasks such as reasoning segmentation, where a model must generate a mask in response to a complex text query (rather than simply a class label or bounding box). For example, *LISA* [8] introduces a special token `[SEG]` and uses an embedding-as-mask paradigm to enable a multimodal LLM to output segmentation masks. However, LISA primarily focuses on generic scenes and reasoning over object categories, and does not explicitly target positional cues or domain-specific vocabularies. More recently, *LISA++* [23] extends LISA with instance differentiation and multi-turn dialogue but still does not specialise in fine positional reasoning or domain adaptation. The work *LLM-Seg* [20] proposes a two-stage pipeline connecting a frozen segmentation foundation model (e.g., Segment Anything Model) and an LLM, and introduces a large "reasoning segmentation" dataset (LLM-Seg40K) to address data scarcity. Meanwhile, *SegLLM* [21] presents a multi-round interactive reasoning segmentation approach that integrates conversational memory and segmentation masks. In parallel, PixelLM [16] aligns pixel-level visual features with linguistic reasoning through token-to-pixel correspondence, bridging dense prediction and language understanding. However, it remains limited in domain-specific reasoning and positional alignment. To address this gap in the medical domain, PRS-Med [19] introduces Positional Reasoning Segmentation for Medical Imaging, integrating vision–language models with segmentation to jointly generate accurate masks and spatial reasoning outputs. Nevertheless, PRS-Med shows these limitations: the generated reasoning text is often not semantically close to the ground-truth explanations, and its reasoning quality is evaluated using a binary Yes/No response from ChatGPT, which is not robust and fails to capture semantic or spatial differences. Our proposed CG-Reasoner addresses these shortcomings by enforcing explicit semantic and spatial grounding.

2.3. Reasoning Metrics for Segmentation

Evaluating reasoning-based segmentation requires metrics that jointly assess linguistic understanding, spatial reasoning, and visual grounding. Traditional natural language metrics such as *BLEU* [14], *ROUGE* [10], and *METEOR* [1] have been widely used to evaluate textual similarity between generated and reference reasoning outputs; however, these measures fail to capture geometric correctness or spatial consistency, both of which are crucial in reasoning-guided segmentation. *PixelLM* [16] introduces a qualitative reasoning evaluation protocol that leverages *GPT-3.5* to score generated reasoning responses from 1 to 10 based on interpretability and contextual correctness, assigning a score of 0 to unrecognized predictions. While effective for open-domain reasoning, this approach is costly, time-consuming, and prone to subjectivity due to dependence on external API-based scoring. In contrast, *PRS-Med* [19] establishes a medical reasoning benchmark by fine-tuning multimodal LLMs—*LLaVA-Med* [9], *HuatuoGPT-Vision* [3], and *MedVLM-R1* [13]—on the MMRS dataset, where reasoning accuracy is evaluated using binary (Yes/No) question–answer tasks. Although this evaluation is domain-specific and interpretable, it remains coarse-grained and requires additional model training. To overcome these limitations, we propose a that jointly evaluates spatial and semantic alignment between reasoning text and segmentation masks.

3. Methods

3.1. Overall Architecture

The proposed CG-Reasoner framework, as shown in Fig. 2, unifies vision and language reasoning through four integrated components: a vision encoder, a multimodal large language model (LLM), a light-weight decoder, and a Text2Centroid module. The vision encoder extracts hierarchical anatomical representations, while the LLaVA-Med model serves as the multimodal LLM that interprets clinical text prompts to generate semantic embeddings reflecting spatial intent. These embeddings guide the light-weight decoder, which fuses linguistic and visual features via cross-modal attention to produce context-aware and anatomically precise segmentation masks. The Text2Centroid module further bridges reasoning and geometry by predicting lesion centroids from textual explanations, reinforcing spatial consistency between language and visual localization.

To incorporate deeper semantic understanding and contextual reasoning, LLaVA-Med is equipped with a lightweight LoRA adapter. The model processes question–answer (QA) pairs describing spatial positions of tumors or other anatomical structures, and through prompt conditioning, generates spatially grounded text embeddings that act as semantic prompts for the decoder. The LoRA adapter fine-tunes only a small subset of parameters, allowing efficient adaptation to medical language without retraining the full LLM. Collectively, these components enable CG-Reasoner to perform interpretable segmentation and positional reasoning, yielding both quantitative (mask-based) and qualitative (language-based) outputs.

3.2. Light-weight Vision Encoder

To achieve semantically rich and spatially precise visual representations, we design a novel Light-weight Vision Encoder, as illustrated in Fig. 2, built upon ConvNeXt-Tiny [11] as the foundational backbone. While ConvNeXt-Tiny provides strong hierarchical representations, our encoder extends it with three key enhancements—feature refinement, multi-scale fusion, and channel attention—to better capture fine structural details and improve adaptation across medical imaging modalities. Let R_i denotes the refined feature map obtained from the ith selected ConvNeXt stage ($i \in \{1, 2, 3\}$), where the refinement is performed using convolution–batch normalization–ReLU blocks to recover subtle boundaries and anatomical textures that are often lost in deeper layers. These refined representations are then bilinearly upsampled and fused residually as:

$$F_{\text{fused}} = R_1 + \text{Up}(R_2) + \text{Up}(R_3), \tag{1}$$

where F_{fused} denotes the fused multi-scale representation and $\text{Up}(\cdot)$ represents bilinear interpolation for spatial alignment. This residual fusion effectively integrates both local structural cues and global semantic context, enhancing lesion localization and cross-modality robustness. Finally, an adaptive channel reweighting stage is applied to the fused feature map, where global average pooling followed by nonlinear transformations selectively emphasizes informative feature channels and suppresses redundant ones, yielding boundary-aware and high-fidelity visual embeddings suitable for prompt-guided segmentation and positional reasoning.

3.3. Light-weight Decoder

The Decoder, as shown in Fig. 2, integrates the visual encoder's representations with language-derived prompts to generate segmentation masks consistent with textual intent. It comprises three main components: (i) *prompt projection*, (ii) *FiLM-based conditioning*, and (iii) *lightweight fusion and upsampling*.

Given prompt embeddings $\mathbf{P} \in \mathbb{R}^{B\times T\times d_p}$ from the LLM, where B denotes the batch size, T is the number of tokens in the prompt, and d_p is the dimensionality of the prompt embeddings, we project them into the same feature space as the visual encoder features with dimension d_v. Specifically, for each token embedding $\mathbf{P}_t \in \mathbb{R}^{B\times d_p}$, we compute

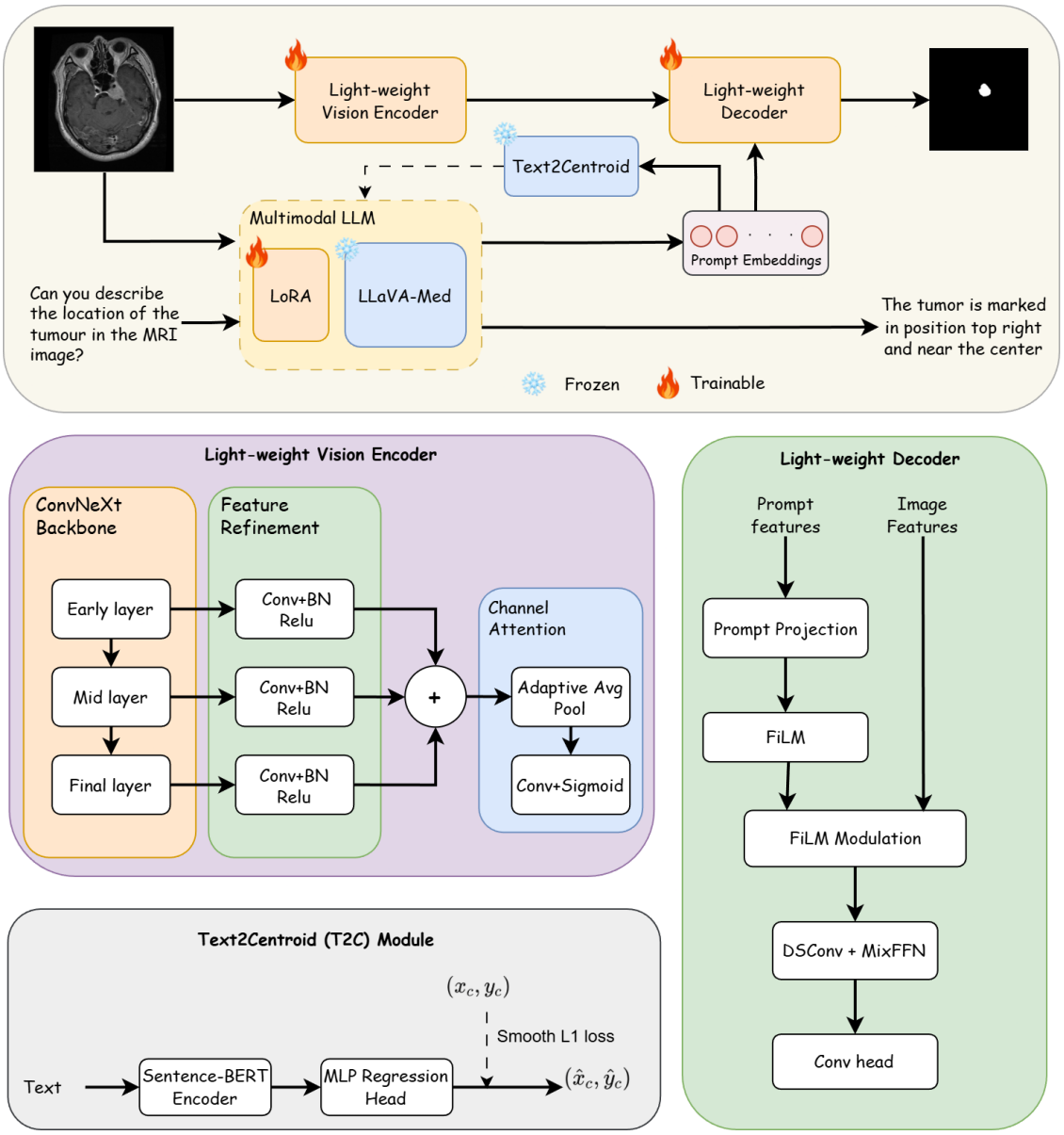


Figure 2. **Overview of the proposed CG-Reasoner framework.** The top figure illustrates the overall architecture comprising four key components—*Light-weight Vision Encoder*, *Multimodal LLM (LLaVA-Med + LoRA)*, *Text2Centroid Module*, and *Light-weight Decoder*. The remaining figures depict the detailed architecture of each individual component within the CG-Reasoner framework. The vision encoder extracts hierarchical anatomical features from medical images, while the multimodal LLM processes question–answer (QA) prompts. The Light-weight Decoder fuses these linguistic and visual embeddings, and the Text2Centroid module bridges reasoning and geometry by predicting lesion centroids for spatial alignment.

$$\mathbf{z}_p = \frac{1}{T}\sum_{t=1}^{T} \phi(\mathrm{LN}(\mathbf{P}_t \mathbf{W}_p)), \tag{2}$$

where $\mathbf{W}_p \in \mathbb{R}^{d_p \times d_v}$ is a learnable projection matrix that maps the prompt embeddings into the visual feature space, $\mathrm{LN}(\cdot)$ denotes Layer Normalization, and $\phi(\cdot)$ represents the GELU activation function. The resulting vector $\mathbf{z}_p$ denotes the aggregated prompt representation obtained by averaging the projected token features across all T tokens.

To inject the linguistic context into the visual stream, the decoder applies Feature-wise Linear Modulation (FiLM) [15]. A lightweight MLP transforms the prompt descriptor into channel-wise scaling and bias parameters. The modulated features are refined through a depthwise-separable convolution followed by a MixFFN block [22] for spatial

interaction. The resulting features are hierarchically upsampled through transposed convolutions, and a lightweight convolutional head with sigmoid activation produces the final segmentation mask. Overall, this lightweight decoder tightly couples linguistic intent with visual information, enabling precise, context-aware, and computationally efficient mask generation.

3.4. Text2Centroid (T2C) Module

To evaluate and transfer spatial reasoning into multimodal alignment, we design a novel efficient Text2Centroid (T2C) module, a lightweight network that learns to map descriptive reasoning text into normalized 2D spatial coordinates. During pretraining, each reasoning sentence T_i is paired with its corresponding segmentation mask M_i, whose centroid serves as the spatial supervision signal. Its coordinates (x_c, y_c) are linearly normalized to $[-1, 1]$ such that the top–left corresponds to $(-1, 1)$ and the bottom–right to $(1, -1)$. As shown in Fig. 2, the model consists of a frozen Sentence-BERT encoder followed by a multi-layer regression head. Let N denote the number of training samples and i index each text–mask pair. The model is optimized with a Smooth L1 loss (S_{L1}) between the predicted $(\hat{x}_c, \hat{y}_c)$ and ground-truth centroids:

$$\mathcal{L}_{\text{T2C}} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{S}_{\text{L1}}((\hat{x}_{c,i}, \hat{y}_{c,i}), (x_{c,i}, y_{c,i}))\,, \qquad (3)$$

Once trained, the T2C module's weights are frozen and integrated into the overall multimodal framework, as illustrated in Fig. 2. This module is computationally lightweight—requiring only 10 minutes to train for 10 epochs on a single NVIDIA A100 GPU—yet provides highly effective spatial grounding within the system. Given a predicted reasoning text and its corresponding reference, the pretrained module generates their spatial embeddings, denoted as the ground-truth and predicted positional vectors. The similarity between these vectors serves as the spatial-reasoning consistency score, computed as per Eq. 4, which measures how well the textual reasoning aligns with the underlying visual localization cues (for example, phrases such as "top-right" or "bottom-left"). During end-to-end fine-tuning, this spatial alignment signal backpropagates into the multimodal trainable text encoder (LoRA + LLaVA), enabling the model to learn reasoning that is geometrically grounded in the visual domain.

3.5. PRScore: Positional-Reasoning Score

We introduce PRScore, a novel evaluation metric designed to measure holistic cross-modal consistency in multimodal reasoning with a focus on segmentation. Unlike conventional linguistic metrics such as BLEU, ROUGE, or METEOR that focus solely on lexical overlap, PRScore, as illustrated in Fig. 3, jointly assesses three complementary dimensions—*semantic fidelity*, *spatial reasoning*, and *visual grounding*—providing a unified measure of alignment between language, reasoning, and visual localization. Each component is normalized within the range [0,1]. For any vectors $\mathbf{u}, \mathbf{v}$, cosine and normalized cosine similarities are defined as:

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}, \qquad \text{ncos}(\mathbf{u}, \mathbf{v}) = \tfrac{1}{2}(\cos(\mathbf{u}, \mathbf{v}) + 1)\,,$$

ensuring $\text{ncos}(\cdot, \cdot) \in [0, 1]$. We compute sentence-level semantic similarity S_{sem} between ground-truth and predicted reasoning texts using SBERT embeddings. To evaluate geometric consistency, we use our pretrained T2C model to map texts into normalized coordinate embeddings. Spatial similarity combines directional and distance alignment:

$$S_{\text{spatial}} = \tfrac{1}{2}\big[\,\text{ncos}(\mathbf{p}_{\text{gt}}, \mathbf{p}_{\text{pred}}) + (1 - d(\mathbf{p}_{\text{gt}}, \mathbf{p}_{\text{pred}}))\big], \quad (4)$$

where $d(\cdot, \cdot)$ is the Euclidean distance between 2D coordinates normalized to $[0, 1]$. The text-mask alignment for visual grounding S_{vis} is also similarly defined as

$$S_{\text{vis}} = \tfrac{1}{2}\big[\,\text{ncos}(\mathbf{p}_{\text{pred}}, \mathbf{m}) + (1 - d(\mathbf{p}_{\text{pred}}, \mathbf{m}))\big], \qquad (5)$$

where $\mathbf{m}$ refers to centroid of the generated mask. Finally, the normalized PRScore is computed as a weighted combination:

$$\text{PRScore} = \alpha\, S_{\text{sem}} + \beta\, S_{\text{spatial}} + \gamma\, S_{\text{vis}}, \qquad \alpha + \beta + \gamma = 1, \quad (6)$$

where equal weights $\alpha = \beta = \gamma = \frac{1}{3}$ are used to avoid biasing the metric towards any single modality. The score ranges within [0,1], with higher values indicating stronger multimodal alignment in meaning, reasoning, and localization.

3.6. Need for a Novel PRScore Metric

Evaluating models that generate both segmentation masks and reasoning text requires a metric that jointly measures the semantic and spatial consistency between the textual explanation and the corresponding visual evidence. Traditional rule-based metrics such as ROUGE, BLEU, and METEOR depend on exact n-gram or token-level overlaps between generated and reference texts. However, these metrics are brittle and fail to capture semantic or spatial intent. They cannot generalize to diverse phrasing, paraphrases, or context-dependent reasoning statements that convey equivalent meaning (e.g., "upper-left" and "top-left"), and they lack sensitivity to linguistic quality or medical terminology. More critically, these conventional metrics do not penalize spatial inaccuracies—for instance, if the ground-truth description states that the lesion is in the *top-left* region while

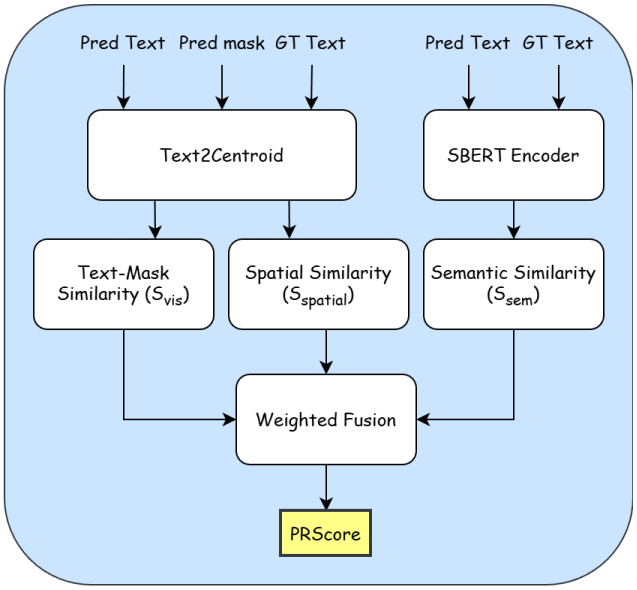


Figure 3. Overview of the proposed framework for computing PRScore.

the predicted text states it is in the *bottom-right*, the score remains artificially high despite the incorrect spatial reasoning. In contrast, the proposed PRScore metric explicitly integrates spatial reasoning by significantly penalizing incorrect positional references in the predicted text, thereby ensuring that semantic and geometric consistency are evaluated together.

While recent LLM-as-a-Judge approaches (e.g., GPTScore) offer semantic flexibility, they introduce subjectivity, inconsistency, and dependency on the evaluator model's prompt or version. To overcome these limitations, PRScore provides an objective, interpretable, and domain-agnostic framework that captures both linguistic fidelity and spatial grounding, ensuring fair evaluation of multimodal segmentation–reasoning models. It is fully automated, reproducible, and computationally efficient—eliminating API latency, subjective bias, and additional training costs—while offering fine-grained insight into spatial reasoning performance. We present a detailed comparison between conventional metrics and PRScore using specific examples of ground-truth text and predicted reasoning text in the supplementary material due to page-limit constraints.

3.7. Loss Function

Our multimodal segmentation–reasoning framework jointly optimizes visual accuracy and spatial interpretability through a unified loss formulation that integrates segmentation, reasoning, and spatial grounding objectives. The overall objective is expressed as

$$\mathcal{L}_{total} = \lambda_{seg}\mathcal{L}_{seg} + \lambda_{txt}\mathcal{L}_{txt} + \lambda_{spatial}\mathcal{L}_{spatial}, \quad (7)$$

where λ_{seg}, λ_{txt}, and $\lambda_{spatial}$ control the contribution of each term. The segmentation loss $\mathcal{L}_{seg}$ combines Dice and Binary Cross-Entropy (BCE) components to ensure fine-grained tumor boundary precision, while the reasoning loss $\mathcal{L}_{txt}$ uses cross-entropy over generated and reference QA tokens to encourage clinically coherent textual outputs. Most importantly, the Spatial Grounding Loss $\mathcal{L}_{spatial}$, derived from a frozen *Text2Centroid* module, enforces geometric alignment between the LLM's linguistic reasoning and the spatial localization inferred by the segmentation mask. This loss directly supervises the LoRA adapter within the *LLaVA-Med* model, enabling it to learn spatially grounded representations that link textual descriptions with precise anatomical regions, thereby ensuring interpretable, position-aware reasoning in medical image understanding.

4. Results and Experiments

4.1. Dataset

We conduct our experiments using six datasets across different imaging modalities. Each sample in the dataset couples a medical image with natural language prompts that describe the spatial location of a tumour or region of interest, enabling cross-modal reasoning during segmentation. The combined dataset [19] spans six major imaging modalities—CT, MRI, X-ray, ultrasound, endoscopy, and RGB.

4.2. Implementation Details

We conduct all experiments using 2 NVIDIA A100 80GB GPUs. CG-Reasoner is trained for 20 epochs with a batch size of 4, requiring approximately two days to complete. We use the AdamW optimizer with a learning rate of 1×10^{-4}, and the best-performing checkpoint is selected based on validation performance, typically around epoch 18. For the multimodal LLM, we apply LoRA with a rank of 16 and an identical value for α, along with a dropout rate of 0.05. All LoRA weights are initialized using a uniform distribution. To ensure fair comparison, we finetuned all the baseline models on the same dataset using identical training parameters.

4.3. Segmentation Results

As shown in Tables 1 and 2, the proposed CG-Reasoner achieves superior segmentation performance across six medical imaging modalities, despite employing a *lightweight encoder–decoder architecture*. Compared to strong multimodal reasoning and foundation models such as G-Dino + SAM-Med2D [4], Biomedparse [24], and LISA-7B/13B [8], CG-Reasoner consistently improves both mean Dice (mDice) and mean Intersection-over-Union (mIoU) scores. Notably, CG-Reasoner attains an mDice of 0.977 and mIoU of 0.958 on the *Lung X-ray* dataset—the highest across all baselines. It also records strong results on *Lung*

Table 1. **Quantitative segmentation results (Part 1):** Comparison across Breast Ultrasound, Brain MRI, and Lung CT-Scan datasets using mDice and mIoU. Higher values indicate better segmentation performance.

Method	Breast Ultrasound		Brain MRI		Lung CT-Scan	
	mDice ↑	mIoU ↑	mDice ↑	mIoU ↑	mDice ↑	mIoU ↑
G-Dino + SAM-Med2D [4]	0.515	0.441	0.667	0.625	0.540	0.392
Biomedparse [24]	0.783	0.698	0.294	0.245	0.088	0.061
LISA-7B [8]	0.299	0.246	0.478	0.402	0.478	0.402
LISA-13B [8]	0.526	0.441	0.124	0.076	0.656	0.528
PRS-Med [19]	**0.817**	**0.729**	0.803	**0.757**	0.968	0.943
PixelLM [16]	0.796	0.692	0.619	0.495	0.946	0.903
CG-Reasoner (Ours)	0.765	0.669	**0.819**	0.731	**0.970**	**0.948**

Table 2. **Quantitative segmentation results (Part 2):** Comparison across Lung X-ray, Polyp Endoscopy, and Skin RGB Image datasets using mDice and mIoU. Higher values indicate better segmentation performance.

Method	Lung X-ray		Polyp Endoscopy		Skin RGB Image	
	mDice ↑	mIoU ↑	mDice ↑	mIoU ↑	mDice ↑	mIoU ↑
G-Dino + SAM-Med2D [4]	0.401	0.300	0.488	0.418	0.237	0.171
Biomedparse [24]	0.972	0.949	0.824	0.774	0.924	0.867
LISA-7B [8]	0.179	0.122	0.241	0.202	0.464	0.368
LISA-13B [8]	0.664	0.535	0.312	0.247	0.643	0.536
PRS-Med [19]	0.969	0.942	**0.843**	**0.791**	0.875	0.799
PixelLM [16]	0.560	0.392	0.699	0.600	0.666	0.539
CG-Reasoner (Ours)	**0.977**	**0.958**	0.716	0.636	**0.904**	**0.840**

CT-Scan (mDice: 0.970, mIoU: 0.948), *Brain MRI* (mDice: 0.819, mIoU: 0.731), and *Skin RGB Image* (mDice: 0.904, mIoU: 0.840). Only marginally lower values appear on *Breast Ultrasound* and *Polyp Endoscopy* datasets, emphasizing that performance drops are minimal. These outcomes highlight that CG-Reasoner not only excels in segmentation accuracy but also generalizes effectively across diverse modalities, balancing reasoning ability with computational efficiency. By leveraging context-guided multimodal reasoning and cross-granularity feature fusion, CG-Reasoner sets a new standard for interpretable multimodal medical segmentation, outperforming heavier models while remaining computationally lightweight. Due to page-limit restrictions, a visual comparison of the segmentation results is presented in the supplementary material.

4.4. Reasoning Results

Table 3 presents the reasoning performance across six medical imaging modalities, where a higher PRScore reflects not only stronger multimodal alignment between linguistic reasoning and spatial localization, but also greater semantic proximity between the generated reasoning text and the ground-truth clinical description. Across all modalities, our CG-Reasoner not only surpasses PRS-Med in four out of six datasets—Breast Ultrasound (0.755), Brain MRI (0.722), Lung CT-Scan (0.847), and Lung X-ray (0.728)—but also consistently outperforms LISA-7B by a large margin in every modality. Although PRS-Med performs slightly better in Polyp Endoscopy (0.734 vs. 0.732) and Skin RGB (0.735 vs. 0.730), CG-Reasoner demonstrates overall superior cross-modal reasoning capability. This consistent improvement is attributed to the LLaVA-Med model trained with a spatial loss derived from the Text2Centroid module, which enforces geometric alignment between linguistic reasoning and visual localization. By integrating spatially grounded supervision, CG-Reasoner effectively bridges the gap between semantic understanding and spatial awareness, resulting in interpretable and position-aware reasoning across diverse medical modalities.

4.5. Ablation Studies

4.5.1. Why ConvNeXt-Tiny is our Vision Backbone

Tables 5 and 6 summarize the segmentation results of several state-of-the-art (SOTA) vision encoders across six medical imaging modalities. Among all competing architectures, ConvNeXt-Tiny demonstrates consistently strong performance with highest mDice and mIoU scores. We adopt ConvNeXt-Tiny [11] as our vision backbone because it offers an optimal balance between accuracy and effi-

Table 3. **Quantitative reasoning evaluation across six medical imaging modalities using the proposed PRScore metric (higher is better).** CG-Reasoner achieves the highest scores across most modalities, highlighting its ability to align reasoning text with ground-truth descriptions while maintaining spatial consistency.

Method	Breast Ultrasound	Brain MRI	Lung CT	Lung X-ray	Polyp Endoscopy	Skin RGB
LISA-7B [8]	0.361	0.406	0.694	0.356	0.304	0.236
PRS-Med [19]	0.729	0.712	0.830	0.697	**0.734**	**0.735**
CG-Reasoner (Ours)	**0.755**	**0.722**	**0.847**	**0.728**	0.732	0.730

Table 4. **Effect of the Text2Centroid Module:** Quantitative comparison of CG-Reasoner with and without the Text2Centroid module. Incorporating this component improves multimodal reasoning by enhancing spatial grounding and semantic alignment across all imaging modalities (higher PRScore is better).

Configuration	Breast Ultrasound	Brain MRI	Lung CT	Lung X-ray	Polyp Endoscopy	Skin RGB
Without Text2Centroid	0.721	0.705	0.832	0.700	0.723	0.719
With Text2Centroid (Ours)	**0.755**	**0.722**	**0.847**	**0.728**	**0.732**	**0.730**

Table 5. **Segmentation performance of SOTA encoders (Part 1):** Comparison on Breast Ultrasound, Brain MRI, and Lung CT-Scan datasets using mDice and mIoU.

Method	**Params (M)**	Breast		Brain		Lung CT	
		mDice	mIoU	mDice	mIoU	mDice	mIoU
Tiny-SAM [18]	31.5	0.601	0.482	0.624	0.510	0.960	0.923
Med-SAM [12]	105.0	0.672	0.520	0.712	0.598	0.920	0.915
Swin-UNet [2]	49.0	0.560	0.385	0.594	0.480	0.883	0.864
ConvNeXt-Tiny [11]	51.0	**0.738**	**0.630**	**0.796**	**0.695**	**0.961**	**0.930**

Table 6. **Segmentation performance of SOTA encoders (Part 2):** Comparison on Lung X-ray, Polyp Endoscopy, and Skin RGB datasets using mDice and mIoU.

Method	Lung X-ray		Polyp		Skin	
	mDice	mIoU	mDice	mIoU	mDice	mIoU
Tiny-SAM [18]	0.971	0.941	0.645	0.400	0.689	0.582
Med-SAM [12]	0.965	0.933	0.630	0.423	0.710	0.684
Swin-UNet [2]	0.920	0.896	0.622	0.364	0.675	0.523
ConvNeXt-Tiny [11]	**0.977**	**0.957**	**0.705**	**0.615**	**0.888**	**0.813**

ciency. As a modernized convolutional architecture incorporating design principles from transformers, ConvNeXt-Tiny employs large-kernel depthwise convolutions, inverted bottlenecks, and LayerNorm-based normalization to produce highly expressive feature maps. These features enable the extraction of rich multi-scale representations of anatomical structures, making it particularly effective for tumor boundary delineation and fine-grained medical segmentation tasks. Its strong performance across all modalities in Tables 5–6 reaffirms that it is an ideal backbone choice for our cross-modal reasoning framework, ensuring high accuracy while preserving modest computational efficiency.

4.5.2. Influence of the Text2Centroid Module

To evaluate the contribution of the Text2Centroid module, we compare the reasoning performance of CG-Reasoner trained with and without this component. By providing explicit spatial grounding—predicting lesion centroids from textual explanations—it guides the LLM to associate linguistic reasoning with geometric localization. Removing this spatial signal causes the model to produce reasoning that may remain semantically correct but becomes notably less precise in spatial terms. As shown in Table 4, integrating this module yields consistent improvements across all six datasets. These results demonstrate that spatial grounding significantly enhances the model's ability to capture anatomical relationships, resulting in more position-aware and clinically interpretable explanations.

5. Conclusion

In this work, we present CG-Reasoner, a unified cross-modal framework that integrates a multimodal LLM, a new light-weight encoder–decoder architecture, and a novel Text2Centroid module to jointly perform medical image segmentation and positional reasoning. Alongside this framework, we introduce PRScore, a novel metric that evaluates both semantic fidelity and spatial alignment between reasoning text and segmentation masks. Experiments across six imaging modalities show that CG-Reasoner achieves state-of-the-art performance, producing accurate segmentation, spatially grounded reasoning, and clinically meaningful explanations within a computationally efficient design. As part of future work, we focus on extending CG-Reasoner to describe lesions in relation to surrounding anatomical structures, enabling richer and more clinically contextualized reasoning for medical report generation.

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